Erfan Shafagh

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Burnaby, BC, Canada

Research Interests

Machine learning, large language models, software security, LLM-based vulnerability detection, commit-level vulnerability detection, automated vulnerability patching, agentic graph-based program analysis, adversarial machine learning, fraud detection, weak supervision, anomaly detection.

Education

May 2026 – Present **Simon Fraser University** – Burnaby, BC
Master of Computer Science. *GPA: 4.0.*

Jan 2023 – Dec 2025 **Simon Fraser University** – Burnaby, BC
Bachelor of Computer Science. *GPA: 3.46.*

Publications

2025 **CleverCatch: A Knowledge-Guided Weak Supervision Model for Fraud Detection**
A. Mozafari, K. Hashemi, **E. Shafagh**, S. Motamedi, A. Taheri, and M. A. Tayebi.
IEEE International Conference on Big Data. arxiv.org/abs/2510.13205

Research & Professional Experience

Nov 2024 – Present **Research Assistant – Simon Fraser University, TBLab**
Conducted research on fraud detection using weak supervision and anomaly detection, contributing to the full cycle from problem formulation through experimentation and publication in collaboration with an interdisciplinary team.

May 2026 – Present **AI Engineer Intern – Farpoint Technologies**
Contributed to Fabric, a secure and private AI assistant built as an agentic harness around large language models, designing and shipping core product features such as streaming reasoning components, onboarding flows, and file search, and resolving cross-platform bugs spanning encoding, rendering, and WSL environments.

Selected Projects

Jul – Dec 2025 **Agent-Based Adversarial & Defensive Learning for Web Application Firewalls**
Fine-tuned multiple LLMs using reinforcement learning (GRPO) to generate adversarial payloads (SQLi, XSS) capable of bypassing WAFs, with custom reward functions optimizing payload diversity and effectiveness.

Sep – Nov 2025 **Corridor Robot – MARS Lab**
Built a real-time path planning system in ROS2 for an RB1-base robot using Model Predictive Path Integral (MPPI) control and JAX-accelerated parallel computation for safe human avoidance in dynamic corridors.

Jan – Apr 2025 **Autonomous Cleaning Robot Path Planning – MARS Lab**
Implemented a Monte Carlo Tree Search planner in ROS2 for a TurtleBot3, using a grid-based environment representation and an optimized reward system balancing coverage efficiency, collision avoidance, and path smoothness.

Feb – Apr 2025 **AI Song Lyric Generation**
Fine-tuned GPT-2 for genre-specific lyric generation, comparing parameter-efficient techniques (Residual Adapters and LoRA) and evaluating outputs via perplexity metrics and human assessment of lyrical quality.

Apr 2025 **Data-Driven Urban Exploration Tools**
Built a location-based service combining Haversine-distance filtering, OpenRouteService route optimization, and a virtual tour-guide feature that identifies landmarks along interpolated paths.

Sep – Nov 2024 **Cinepass**
Collaborated in a team of six to deliver a full-stack movie reservation web application using Spring Boot and PostgreSQL for secure authentication and bookings, following Agile development cycles.

Apr 2024 **Database Management System**
Designed and implemented a normalized relational database in SQLite, applying ER modeling, schema creation, data population, and complex query design to efficiently manage large volumes of structured data.

Feb 2024 **Socket Networking Chat Program**
Developed a multithreaded UDP socket chat application in C enabling two-way communication over a network, with concurrent operations handled to ensure responsiveness and reliable message delivery.

Jun 2023 **Spotify API Integration**
Wrote a Python script to extract metadata from audio files across multiple formats (MP3, FLAC, M4A) and automatically populate a Spotify playlist via the Spotify Web API.

Teaching & Service

May 2026 – Present **Teaching Assistant – Simon Fraser University, CMPT 125**
Led lab sections and office hours guiding students through core programming concepts, data structures, and algorithm design in C, and graded assignments and exams with detailed feedback to strengthen students' problem-solving skills.

Mar 2026 – Present **Program Coordinator – FOSINT-SI Symposium**
Managed paper submissions and author–reviewer communications, and designed and maintained the conference website and media content.

Feb – Mar 2023 **Python Tutor (Volunteer) – Simon Fraser University**
Provided individualized guidance to an undergraduate student on Python fundamentals, adapting teaching approaches to the student's learning style and pace.

Honors & Awards

Spring 2024 President's Honour Roll – Simon Fraser University

Spring 2025 & Summer 2024	Dean's Honour Roll – Simon Fraser University

Technical Skills

Languages	Python, C++, C, Java
ML / DL	PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face, JAX, Kaggle, Google Colab
Databases	SQL, PostgreSQL, SQLite
Systems & Tools	Linux, Git, GitHub, Docker, ROS2, Spring Boot